

6. Crude oil can be separated into simpler mixtures, called fractions, which contain hydrocarbon compounds with boiling points in a similar range.

(a) The table lists the properties of some fractions obtained from crude oil.

Fraction	Number of carbon atoms in fraction	Boiling point range (°C)	Colour of fraction	Flame when burning	Ease of burning
fuel oil	1-4	-170 to 20	colourless	clean	very easy
petrol	5-10	20 to 70	pale yellow	clean	easy
naphtha	8-12	70 to 120	yellow	some soot	quite easy
kerosene	10-16	120 to 240	dark yellow	smoky	quite difficult
diesel oil	15-30	240 to 350	brown	smoky	difficult

Use the information in the table to describe how the burning of the fractions depends upon the size of the molecules. [2]

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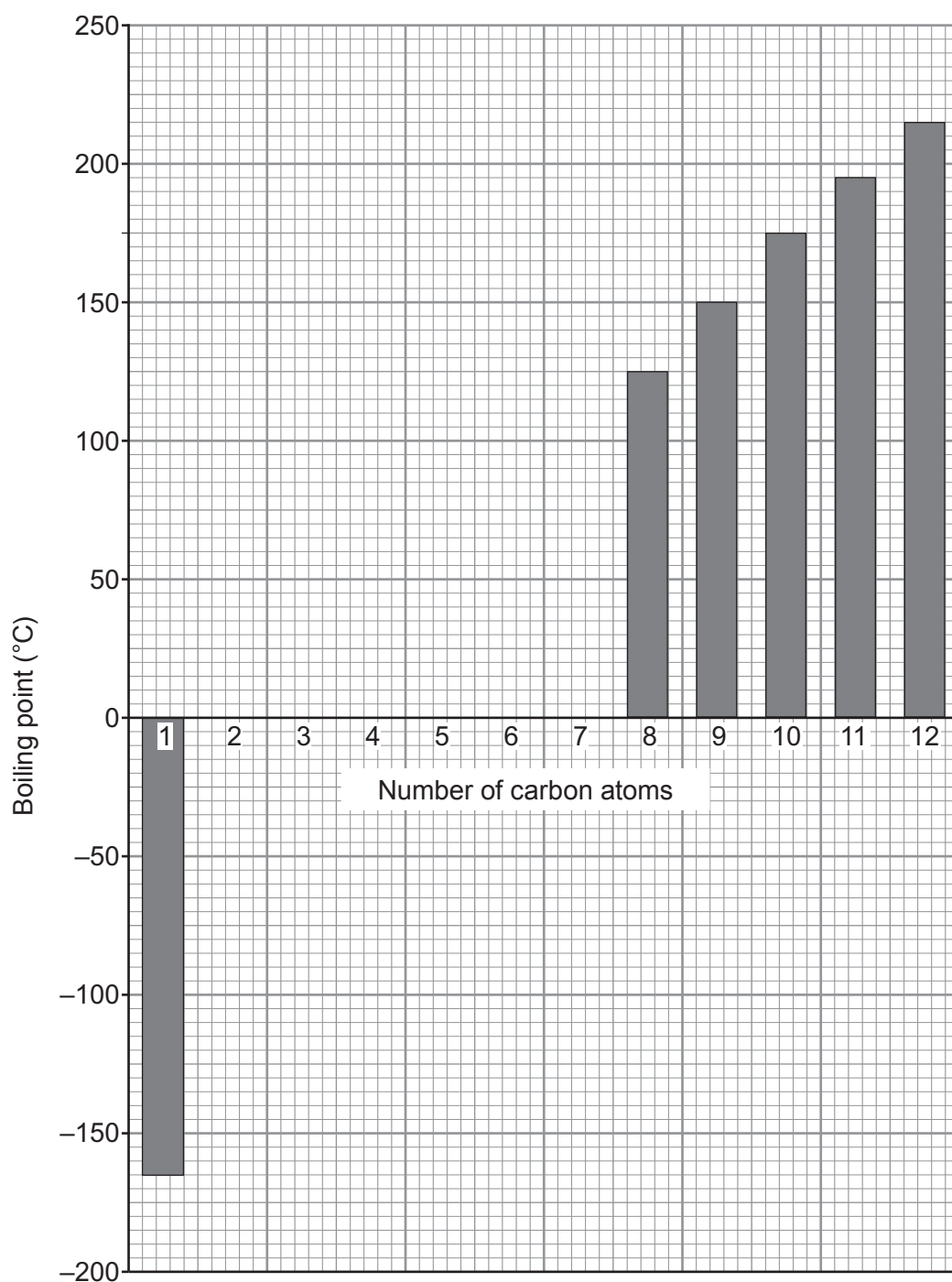
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- (b) The boiling points of hydrocarbons containing 1 to 12 carbon atoms are shown in the table below. The boiling point for the hydrocarbon with 7 carbon atoms is missing.

Number of carbon atoms	Boiling point (°C)
1	-165
2	-90
3	-40
4	10
5	35
6	70
7	
8	125
9	150
10	175
11	195
12	215



- (i) Complete the bar chart below. Some of the bars have already been drawn. [2]



- (ii) Use a ruler to draw a trend line **on the chart** and use this to estimate the boiling point of the hydrocarbon with 7 carbon atoms. [2]

Boiling point °C



(c) Many of the fractions obtained from crude oil are used as fuels.

(i) The fire triangle shows the factors necessary to start and maintain a fire.

State **one** method that could be used to safely put out a small amount of spilled petrol burning on the floor. Give the reason why your chosen method would work. [1]

Method

Reason

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(ii) One of the hydrocarbons in petrol is pentane, C_5H_{12} .

Complete and balance the symbol equation for the complete combustion of pentane. [2]



(iii) Hydrogen fuel cells are now used in many cars instead of petrol. The overall change inside a hydrogen fuel cell is the same as when hydrogen burns.

Explain why using hydrogen fuel cells in cars is better for the environment than petrol. [2]

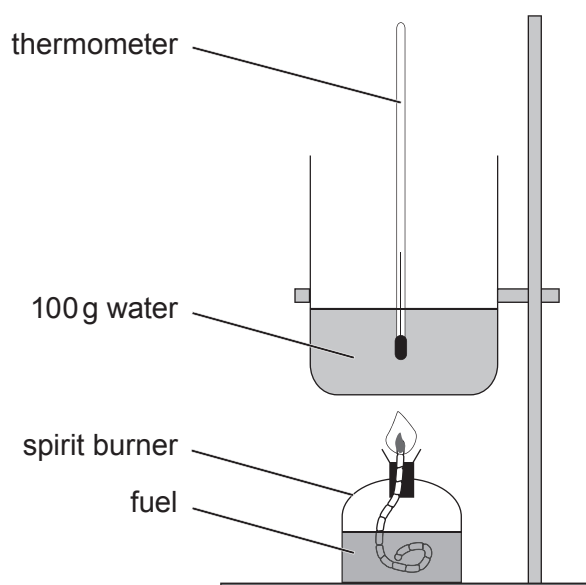
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- (d) It is possible to compare the energy released when different fuels are burned using the following apparatus.



To calculate the energy released per gram of fuel burned, the following equation is used.

$$\text{energy released per gram of fuel (J/g)} = \frac{\text{mass of water} \times 4.2 \times \text{temperature rise (}^{\circ}\text{C)}}{\text{mass of fuel used (g)}}$$

- (i) Apart from measuring the mass of water, describe **all** the measurements that would need to be taken to be able to calculate the energy released per gram of fuel burned. [2]

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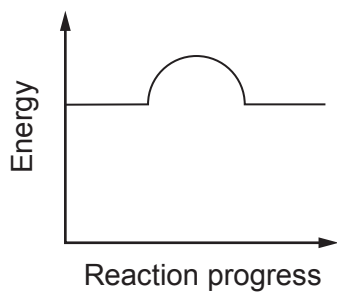
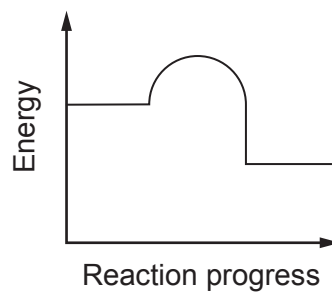
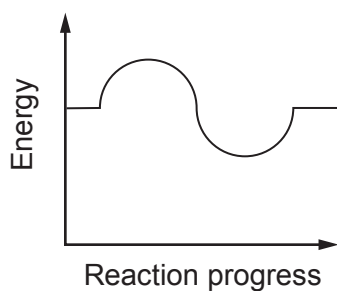
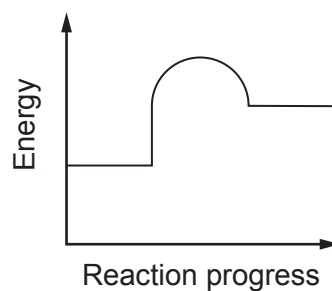
- (ii) When comparing the energy released from different fuels, 100 g of water should be used each time. State **one** other variable that should be controlled. [1]

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(iii) Tick (✓) the energy profile diagram that represents the combustion of fuels. [1]

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END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^{-}
ammonium	NH_4^{+}	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^{-}
calcium	Ca^{2+}	fluoride	F^{-}
copper(II)	Cu^{2+}	hydroxide	OH^{-}
hydrogen	H^{+}	iodide	I^{-}
iron(II)	Fe^{2+}	nitrate	NO_3^{-}
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^{+}	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		



THE PERIODIC TABLE

0

7

6

5

4

3

Group

2

1

28

11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
59 Co Cobalt 27	56 Fe Iron 26	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	
103 Rh Rhodium 45	101 Ru Ruthenium 44	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	
192 Ir Iridium 77	190 Os Osmium 76	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	
186 Re Rhenium 75	184 W Tungsten 74	186 Os Osmium 76	186 Re Rhenium 75	186 Os Osmium 76	
181 Ta Tantalum 73	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Os Osmium 76	
93 Nb Niobium 41	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	101 Ru Ruthenium 44	
45 Sc Scandium 21	48 Ti Titanium 22	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	
55 Mn Manganese 25	56 Fe Iron 26	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	
99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	108 Ag Silver 47	112 Cd Cadmium 48	
106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	
122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54	131 Xe Xenon 54	
127 I Iodine 53	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54	131 Xe Xenon 54	
80 Br Bromine 35	79 Se Selenium 34	75 As Arsenic 33	73 Ge Germanium 32	70 Ga Gallium 31	
40 Ar Argon 18	32 S Sulfur 16	31 P Phosphorus 15	28 Si Silicon 14	27 Al Aluminium 13	
20 Ne Neon 10	16 O Oxygen 8	14 N Nitrogen 7	12 C Carbon 6	11 B Boron 5	
4 He Helium 2					

1
H
Hydrogen
1

7 Li Lithium 3	9 Be Beryllium 4
23 Na Sodium 11	24 Mg Magnesium 12
39 K Potassium 19	40 Ca Calcium 20
86 Rb Rubidium 37	88 Sr Strontium 38
133 Cs Caesium 55	137 Ba Barium 56
223 Fr Francium 87	226 Ra Radium 88
	227 Ac Actinium 89

Key

Ar
Symbol
Name
Z

relative atomic mass

atomic number